

Futureproof

EPM v14a

Parameter Optimisation — Phase 2

Borrower Income Preserved at \$250,000

March 2025

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Executive Summary

This Phase 2 optimisation holds the borrower's income **fixed at \$250,000** (\$25,000/year for 10 years) and optimises all other parameters: loan type (P&I vs IO), collar width, holiday thresholds, profit share percentage and frequency, and FP margin. A total of **277 scenarios** were tested with the top 22 validated at 50,000 Monte Carlo paths.

The recommended configuration reduces PoD from **17.9% to 2.8%** (–15.2pp) while increasing FP revenue by **+174%** from \$823,670 to \$2,257,034. The borrower receives the same \$250,000 annuity income, but under P&I the mortgage amortises to \$0 at maturity (vs \$1.6M outstanding under IO).

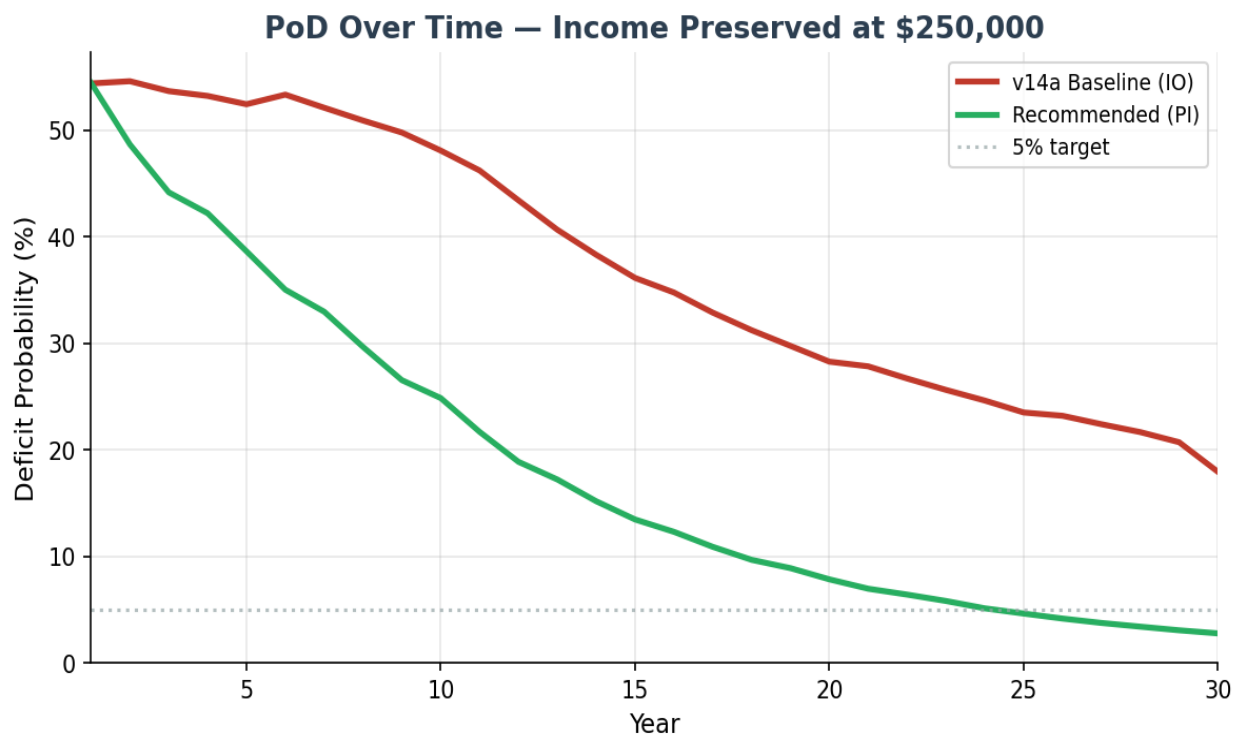
Metric	v14a Baseline	Recommended	Change
Borrower Income	\$250,000	\$250,000	UNCHANGED
PoD Year 30	17.9%	2.8%	-15.2pp
Mean Surplus	\$1,381,981	\$1,755,549	+27%
FP Revenue	\$823,670	\$2,257,034	+174%
Sharpe Ratio	0.829	1.330	+60%
Insurance Premium	\$44,472	\$7,130	-84%
Equity Protection	82.1%	97.2%	+15.1pp
Final Loan Balance	\$1,600,000	\$0	Fully amortised

Recommended Configuration

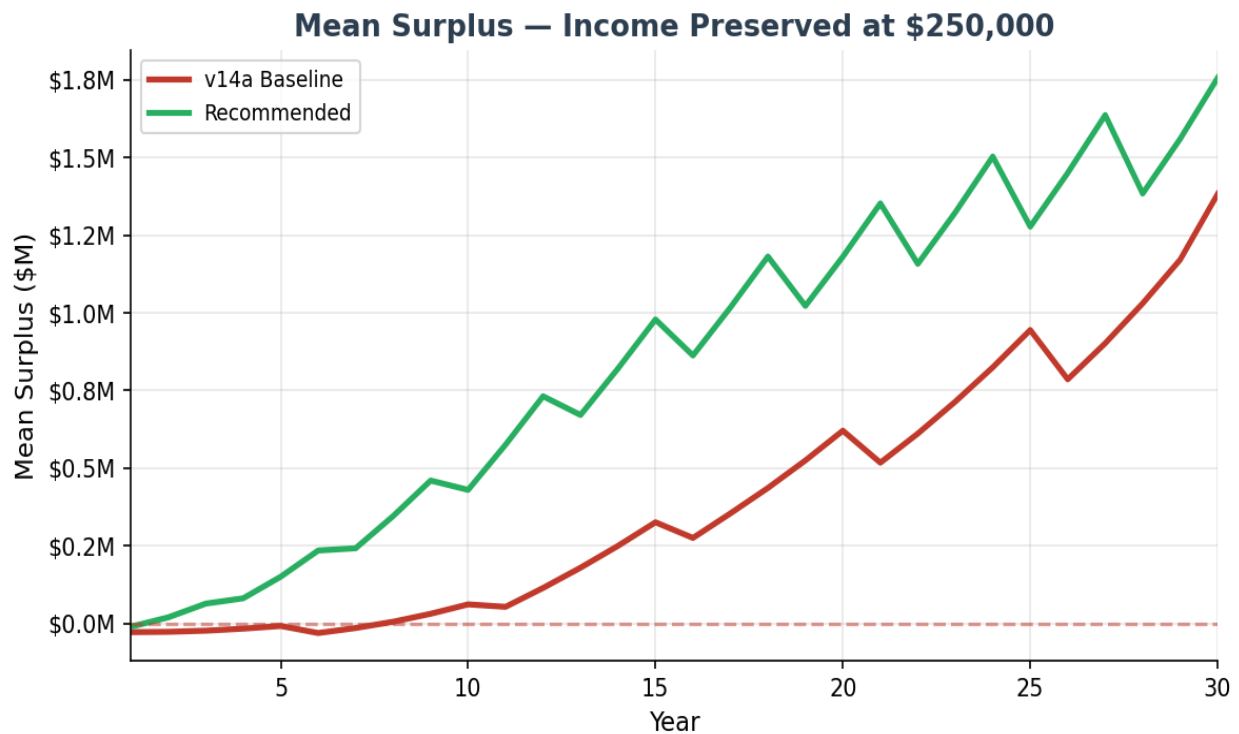
Parameter	v14a Baseline	Recommended	Rationale
Loan Type	IO	PI	Amortises to \$0, eliminates tail risk
Collar Width	±20%	±35%	Captures more equity upside
Holiday Entry	0.90	1.05	More aggressive capital preservation
Profit Share	25% every 5yr	25% every 3yr	More frequent extraction, higher revenue
FP Margin	0.25%	0.15%	Lower margin, better risk metrics
Annuity	\$25K/yr × 10yr	\$25K/yr × 10yr	UNCHANGED

Key changes from v14a baseline: Three structural changes (P&I, wider collar, higher holiday entry) plus two tuning changes (3-year PS frequency, lower FP margin). These work synergistically — P&I reduces the loan balance over time, making the wider collar and tighter holidays even more effective at preventing deficits.

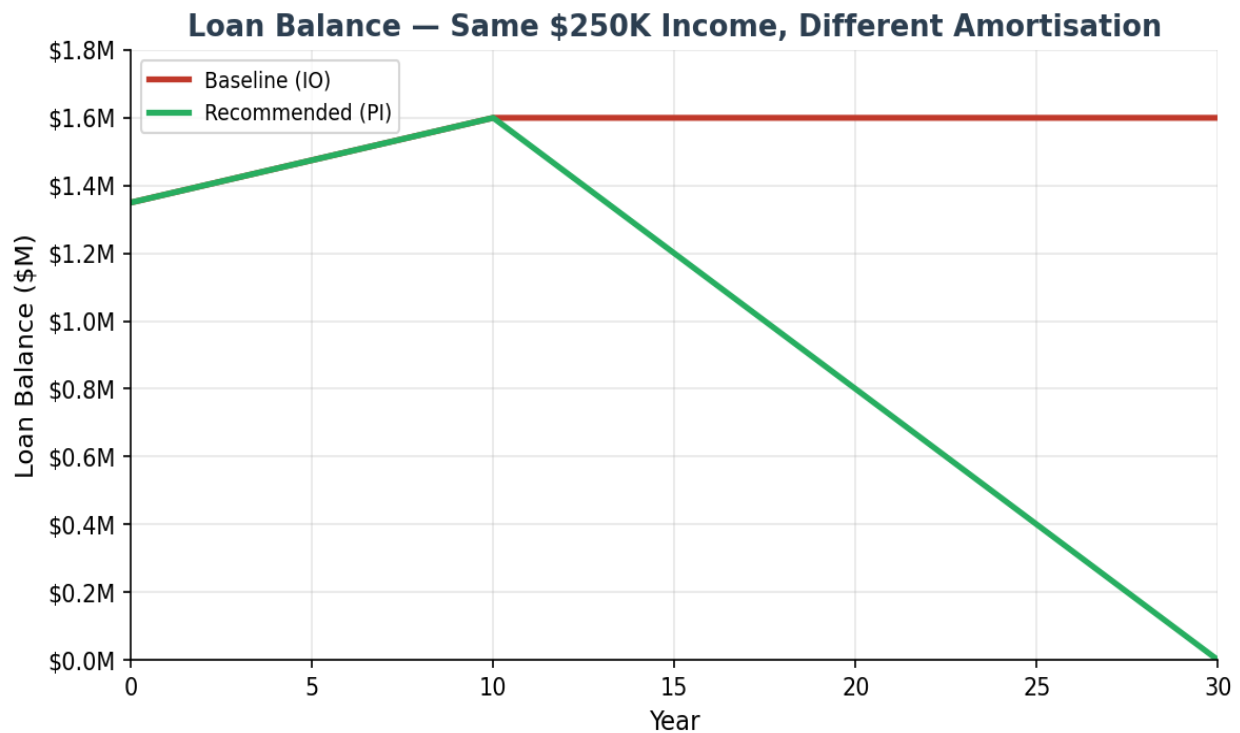
Risk Trajectory



Surplus Trajectory



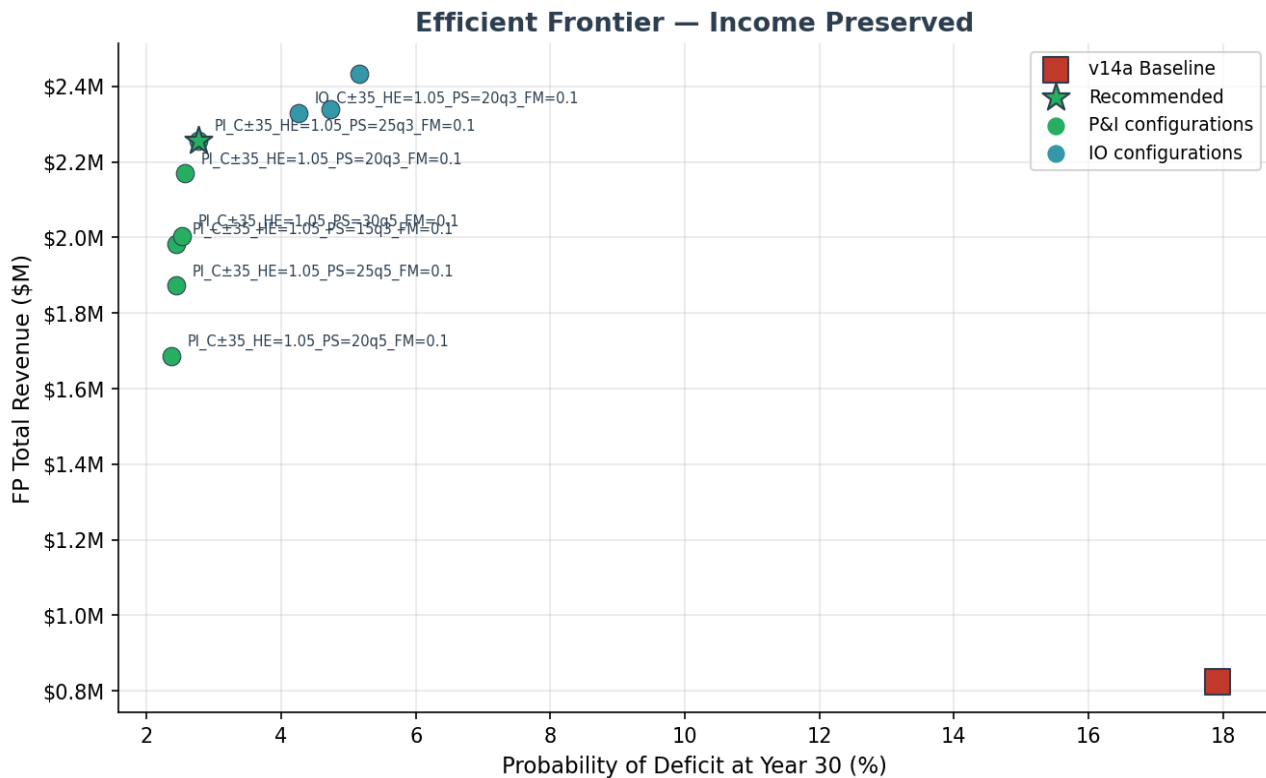
Loan Balance: IO vs P&I;



Both configurations provide the same \$250,000 annuity income over 10 years. Under IO, the \$1.6M loan remains outstanding at maturity. Under P&I, the loan amortises to zero over the remaining 20 years. This fundamental structural change means the investment account must fund ~\$80,000/year in principal repayments after year 10, but the declining loan balance dramatically reduces interest

costs and deficit probability.

Efficient Frontier



The Pareto front contains 9 non-dominated scenarios. All P&I configurations dominate their IO equivalents — for any IO scenario on the front, there exists a P&I scenario with both lower PoD and comparable revenue. The front spans from 2.4% PoD (safest, PI) to 5.2% (highest revenue, IO).

#	Configuration	Type	PoD %	FP Revenue	Sharpe
1	PI_C±35_HE=1.05_PS=20q5_FM=0.1	PI	2.4%	\$1,687,218	1.163
2	PI_C±35_HE=1.05_PS=15q3_FM=0.1	PI	2.4%	\$1,983,284	1.190
3	PI_C±35_HE=1.05_PS=25q5_FM=0.1	PI	2.4%	\$1,872,341	1.201
4	PI_C±35_HE=1.05_PS=30q5_FM=0.1	PI	2.5%	\$2,004,480	1.245
5	PI_C±35_HE=1.05_PS=20q3_FM=0.1	PI	2.6%	\$2,169,672	1.254
6	PI_C±35_HE=1.05_PS=25q3_FM=0.1	PI	2.8%	\$2,257,034	1.330
7	IO_C±35_HE=1.05_PS=20q3_FM=0.1	IO	4.3%	\$2,330,047	1.192
8	IO_C±35_HE=1.05_PS=20q3_FM=0.2	IO	4.7%	\$2,340,588	1.173
9	IO_C±35_HE=1.05_PS=25q3_FM=0.1	IO	5.2%	\$2,433,476	1.206

All Validated Configurations

Top 22 configurations ranked by composite score:

#	Configuration	Type	PoD %	FP Revenue	Sharpe	Mean Surplus
1	PI_C±35_HE=1.05_PS=25q3_FM=0.1 ★	PI	2.8%	\$2,257,034	1.330	\$1,755,549
2	PI_C±35_HE=1.05_PS=20q3_FM=0.1	PI	2.6%	\$2,169,672	1.254	\$2,438,990
3	PI_C±30_HE=1.05_PS=25q3_FM=0.1	PI	2.9%	\$2,141,676	1.361	\$1,653,653
4	PI_C±35_HE=1.05_PS=30q5_FM=0.1	PI	2.5%	\$2,004,480	1.245	\$2,776,966
5	PI_C±35_HE=1.05_PS=15q3_FM=0.1	PI	2.4%	\$1,983,284	1.190	\$3,423,849
6	PI_C±35_HE=1.05_PS=25q3_FM=0.2	PI	3.1%	\$2,253,983	1.315	\$1,699,273
7	PI_C±30_HE=1.05_PS=20q3_FM=0.1	PI	2.8%	\$2,054,573	1.286	\$2,287,327
8	PI_C±35_HE=1.05_PS=20q3_FM=0.2	PI	2.8%	\$2,171,926	1.242	\$2,354,826
9	PI_C±35_HE=1.05_PS=25q5_FM=0.1	PI	2.4%	\$1,872,341	1.201	\$3,446,539
10	PI_C±30_HE=1.05_PS=30q5_FM=0.1	PI	2.7%	\$1,898,499	1.278	\$2,602,871
11	PI_C±30_HE=1.05_PS=25q3_FM=0.2	PI	3.2%	\$2,139,504	1.345	\$1,599,844
12	PI_C±30_HE=1.05_PS=25q5_FM=0.1	PI	2.6%	\$1,771,231	1.231	\$3,222,358
13	PI_C±35_HE=1.05_PS=20q5_FM=0.1	PI	2.4%	\$1,687,218	1.163	\$4,263,920
14	PI_C±25_HE=1.05_PS=25q3_FM=0.1	PI	3.4%	\$1,909,588	1.403	\$1,454,645
15	PI_C±25_HE=1.05_PS=20q3_FM=0.1	PI	3.1%	\$1,823,522	1.332	\$1,991,714
16	PI_C±30_HE=1.05_PS=20q5_FM=0.1	PI	2.5%	\$1,594,547	1.192	\$3,979,317
17	PI_C±25_HE=1.05_PS=25q3_FM=0.2	PI	3.7%	\$1,909,028	1.381	\$1,405,105
18	IO_C±35_HE=1.05_PS=20q3_FM=0.1	IO	4.3%	\$2,330,047	1.192	\$2,923,089
19	IO_C±35_HE=1.05_PS=20q3_FM=0.2	IO	4.7%	\$2,340,588	1.173	\$2,811,668
20	IO_C±35_HE=1.05_PS=25q3_FM=0.1	IO	5.2%	\$2,433,476	1.206	\$2,164,567
21	PI_baseline	PI	10.0%	\$763,481	1.033	\$1,285,615
22	v14a_BASELINE •	IO	17.9%	\$823,670	0.829	\$1,381,981

★ = *Recommended* • = *v14a Baseline*

Detailed Comparison

Metric	v14a Baseline	Recommended
Loan Type	IO	PI
Collar	±20%	±35%
Holiday Entry	0.90	1.05
Profit Share	25% q5	25% q3
FP Margin	0.25%	0.15%
PoD Year 30	17.9%	2.8%
PoD Year 20	28.3%	7.8%
PoD Year 15	36.1%	13.4%
PoD Year 10	48.1%	24.8%
Mean Surplus	\$1,381,981	\$1,755,549
Median Surplus	\$1,111,093	\$1,495,134
P1 Surplus	\$-1,641,957	\$-693,893
P5 Surplus	\$-825,657	\$270,775
P10 Surplus	\$-412,449	\$522,724
P25 Surplus	\$263,920	\$934,312
P75 Surplus	\$2,239,499	\$2,293,833
P90 Surplus	\$3,526,690	\$3,341,854
Std Deviation	\$1,667,641	\$1,320,288
Sharpe Ratio	0.829	1.330
FP Revenue	\$823,670	\$2,257,034
Profit Share	\$672,058	\$2,166,316
FP Margin	\$151,612	\$90,717
Funder Surplus	\$746,484	\$886,671
Insurance Premium	\$44,472	\$7,130
Cond. Exp. Deficit	\$-619,411	\$-645,606
Final Loan Balance	\$1,600,000	\$0
Borrower Protection	82.1%	97.2%

Conclusions

With borrower income held constant at \$250,000, the optimisation identifies a configuration that dramatically improves all risk and revenue metrics:

- **PoD reduction:** From 17.9% to 2.8% (–15.2pp). The borrower's equity is protected in 97.2% of scenarios vs 82.1%.
- **Revenue increase:** FP total revenue rises from \$823,670 to \$2,257,034 (+174%). The 3-year profit share frequency generates more cumulative profit share (\$2,166,316 vs \$672,058).
- **Insurance premium:** Drops 84% from \$44,472 to \$7,130, making the product dramatically cheaper to insure.
- **Loan amortisation:** Under P&I, the mortgage fully amortises to \$0 at maturity. The borrower receives the same \$250,000 income but owns their home outright at the end of the term, vs \$1.6M outstanding under IO.
- **Funder benefit:** Funder surplus share increases from \$746,484 to \$886,671 (+19%), making the product more attractive to wholesale funders.

Implementation

#	Change	Impact	Notes
1	Switch to P&I;	–7.4pp PoD	Structural — amortises to \$0
2	Collar ±35%	–8.5pp PoD	Near-zero collar cost
3	Holiday entry 1.05	–9.7pp PoD	Parameter change only
4	PS 25% every 3yr	+\$183K FP rev	More frequent cash flow
5	FP margin 0.15%	–1.1pp PoD	Lower recurring fee

Methodology

61 + 216 scenarios × 10K paths, top 22 validated at 50K paths. Seed: 42. Constraint: annuity fixed at \$25,000/yr × 10yr = \$250,000.